Problem 4:

a)

$\boldsymbol{A} = \boldsymbol{z}\boldsymbol{z}^{\boldsymbol{T}}$, $z \in \mathbb{R}^n$

To prove that A is positive semidefinite, $\left(x^T A x\right)$ must be greater than or equal to zero.

$$\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{z}\boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x} = \left(\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{z}\right)\left(\boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x}\right) \tag{1}$$

Since, $\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{z} = \boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x} = a$, where $a \in \mathbb{R}$:

$$\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x} = a^2 \tag{2}$$

It can be seen that $\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x}$ is equal to the square of a scalar, hence, $\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x} \geq 0$ and therefore, $\boldsymbol{A}$ is positive semidefinite.

b)

$\boldsymbol{A} = \boldsymbol{z}\boldsymbol{z}^{\boldsymbol{T}}$, $z \in \mathbb{R}^n$

To find the rank of $\boldsymbol{A}$:

$$\boldsymbol{A_i} = z_i \times \boldsymbol{z} \tag{3}$$

Where:

$\boldsymbol{A_i}$: $the\ i_{th}\ column\ of\ \boldsymbol{A}$

z_i: $the\ i_{th} element\ of\ \boldsymbol{z}$

It can be shown from equation (3) that the columns of A are all scalar multiples of the same vector z, therefore, the columns of A are all dependent and the **rank of A is equal to one**.

To find the null space of A:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{z}\boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x} = \boldsymbol{0} \tag{4}$$

$$\left(\boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x}\right)\boldsymbol{z} = \boldsymbol{0} \tag{5}$$

Since $\boldsymbol{z}$ is not a zero vector and $(\boldsymbol{z}^{\boldsymbol{T}}\boldsymbol{x})$ is a scalar:

$$\left(z^T x\right) = 0 \tag{6}$$

This is an inner product between the vector z and the vector x, for the inner product to be zero, x must be orthogonal to z, **hence, the null space of A is the set of all vectors orthogonal to $span(z)$, in other words, the null space of A equals to $span(z)\perp$.**

c)

$A \in \mathbb{R}^{n\times n}$ Is positive semidefinite, $B \in \mathbb{R}^{m\times n}$

Since A is PSD:

$$x^T A x \geq 0 \tag{7}$$

Where:

$x \in \mathbb{R}^n$

To prove that BAB^T is PSD, equation (8) must be true:

$$u^T BAB^T u \geq 0 \tag{8}$$

Where:

$u \in \mathbb{R}^m$

Let: $v = B^T u,\ v \in \mathbb{R}^n$

$$u^T BAB^T u = vAv^T \tag{9}$$

Since v is an arbitrary vector in $\mathbb{R}^n$, from equation (7):

$$u^T BAB^T u = v^T A v \geq 0 \tag{10}$$

Hence, form equation (10), BAB^T is PSD.